

Aditya Singh Bhadauria

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AI/ML Engineer | GenAI, RAG & Agentic Systems | FastAPI | AWS | Document Intelligence

Education

D Y Patil University, Pune

BTech in Artificial Intelligence and Data Science

2022 – 2026

CGPA: 9.49/10

Skills

Programming: Python, SQL

Machine Learning/ Deep Learning: Scikit-learn, XGBoost, PyTorch, TensorFlow, CNNs, ResNet, Model Evaluation, Cross-Validation, Hyperparameter Tuning

GenAI: LangChain, OpenAI APIs, Prompt Engineering, RAG, Embeddings, Agentic AI, Vector Databases

Cloud & DevOps: AWS (S3, VPC, Elastic Beanstalk, Lambda, DynamoDB, Cognito, API Gateway, CloudFront, SageMaker), Docker, Git

Backend & Data: FastAPI, Flask, REST APIs, Pandas, NumPy, EDA, ELT, Data Pipelines, Visualization

System Design: Scalable AI Pipelines, Serverless Architecture, Model Deployment, Evaluation Frameworks

Experience

Cloud Engineering Intern

HeProAI

Dec 2025 – Jan 2026

- Built a serverless task tracker using AWS S3 CloudFront Cognito API Gateway Lambda and DynamoDB following best practices.
- Implemented secure JWT authentication using Cognito User Pools and protected REST APIs.
- Integrated frontend with Lambda backend and DynamoDB enabling full CRUD functionality.

Machine Learning Intern

ALGONIVE

Oct – Nov 2025

- Built a Blood Pressure Prediction Flask web app using Scikit-learn with feature engineering on health attributes (age, weight, BMI, activity level); achieved $R^2 > 0.85$, MAE < 10 mmHg, and $< 2s$ response time.
- Developed ML pipelines for stock analysis and evaluation using regression and time series techniques.
- Serialized trained model using Pickle for single-load-at-startup inference, reducing per-request overhead; built Jinja2-powered Flask UI accessible on low-powered devices.

AWS AI-ML Virtual Internship

AICTE

Apr – Jun 2025

- Built a real-time Eye Blink & Drowsiness Detection system using MediaPipe Face Mesh, OpenCV, and Eye Aspect Ratio (EAR) thresholding; triggered live sound alerts on prolonged eye closure.
- Deployed ML model on Amazon SageMaker with AWS-based inference workflow; implemented scalable pipeline using S3, VPC, and Elastic Beanstalk.

Projects

Document AI – Invoice Extraction System | Tesseract OCR, YOLOv8, Fuzzy Matching, Streamlit

[GitHub](#)

- Built end-to-end invoice extraction pipeline using Tesseract OCR and YOLOv8 for field detection; reduced manual effort by 80% and increased extraction reliability by 30%.
- Achieved 90–95% accuracy via hybrid rule-based and fuzzy matching pipeline; designed confidence scoring and validation system enabling batch processing of 100+ invoices with real-time Streamlit deployment.

AI Medical Report & X-ray Analyzer | ResNet18, Tesseract OCR, Streamlit

[GitHub](#)

- Built multimodal diagnosis system combining Tesseract OCR for report parsing and ResNet18 CNN for image analysis; achieved 85–90% pneumonia detection accuracy and reduced manual review time by 60%.
- Improved diagnostic confidence by 25% via multimodal risk scoring engine; deployed real-time inference dashboard on Streamlit with $< 2s$ end-to-end latency.
- Built a unified inference pipeline combining image and text features to enhance diagnostic accuracy and reliability

AI Stock Trading Agent | FastAPI, React, OpenAI, Upstox API, PostgreSQL

[GitHub](#)

- Built an autonomous NSE investment agent that analyzes 20 stocks using technical indicators and AI-based sentiment to generate BUY HOLD SELL signals with confidence scores.
- Designed a portfolio allocation engine with confidence-based position sizing, dynamic cash reserve based on VIX, and automated stop-loss management.
- Developed a FastAPI backend with PostgreSQL trade logging, SIP automation, and Upstox API integration for paper and live trade execution.